Rethinking Masked Autoencoder for Salient Object Detection in Optical Remote Sensing Images From a Pseudo Image Pretraining Perspective

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***Abstract*— Recently masked autoencoder (MAE) has achieved great success in visual representation learning and delivered promising potential in many downstream vision tasks. However, due to the lack of a saliency supervision signal in the original MAE, almost no saliency information can be learned from the masked image reconstruction process. Therefore, salient object detection (SOD) can hardly benefit from MAE pretraining. To address this issue, we integrate the SOD model and saliency supervision in MAE and propose a simple and effective framework for dynamic hybrid masking MAE (DHMMAE), utilizing MAE with dynamic hybrid masking (DHM) ratios to pretrain SOD model. Specifically, we treat MAE as an online data augmenter to generate endless pseudo images for the SOD model to predict saliency maps, where saliency supervision is employed to ensure that the model can learn robust saliency prior knowledge from the reconstructed images. Besides, we propose a simple and novel network EANet driven by DHMMAE pretraining for SOD in optical remote sensing images (ORSIs). Two key modules are designed for EANet to further enhance its performance: the enhanced diverse feature aggregation module (EDFAM) and the adjacent-context shuffle spatial attention module (ASSAM). EDFAM aggregates diverse features via three different types of convolution layers and enhances them by convolution block attention module (CBAM). ASSAM captures spatial location information of salient objects by employing channel shuffle operation and weighted spatial attention mechanism on the fused adjacent context. Experiments on three ORSI-SOD datasets demonstrate that our proposed method outperforms the cutting-edge methods. Code is available at https://github.com/Voruarn/EANet**

***Index Terms*— Deep learning, masked autoencoder (MAE), optical remote sensing images (ORSIs), salient object detection (SOD).**

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I. Introduction

SALIENT object detection (SOD) is a critical task in computer vision that involves identifying and segmenting the most visually distinctive or significant objects within an image or video sequence [1], [2], [3]. SOD is also a preprocessing step for many computer vision tasks, such as object segmentation [4], [5], object tracking [6], [7], quality assessment [8], [9], and light field segmentation [10], [11]. Recently, SOD in optical remote sensing image (ORSI)-SOD [12], [13], as an emerging topic in understanding ORSI, has received increasing attention, which has been extensively applied in urban planning, traffic monitoring, military reconnaissance, and natural disaster monitoring [14], [15].

In the past few years, with the boom of deep learning, significant advancements have been made in the field of ORSI-SOD. Li et al. [16] designed LVNet and constructed the first dataset (ORSSD) for ORSI-SOD, which is the pioneering work of this community. Later, many researchers followed up and contributed a lot to this field [17], [18], [19], [20]. Zhang et al. [13] proposed DAFNet and the second public dataset (EORSSD) for ORSI-SOD, which boosted the development of this community. Meanwhile, a growing number of researchers devoted themselves to this field, and many excellent works were made, such as RRNet [18], ACCoNet [14], PCDNet [20], BSCGNet [22], and GeleNet [12]. After that, Tu et al. [37] proposed the third-largest dataset (ORSI-4199) for ORSI-SOD, which further pushed forward this field. Nowadays, ORSI-SOD has become a research hotspot in analyzing ORSIs, which has been widely explored.

Over the past few years, from LVNet [16] and DAFNet [13] to MJRBM [37] and GeleNet [12], we have witnessed remarkable advancements in ORSI-SOD [16], [17], [18], [19], [20], [21], [22], [23], [24]. These methods achieve impressive performance gain by introducing novel sophisticated model designs, such as two-stream pyramid architecture [16], dense attention fluid [13], multicontent complementation [17], adjacent-context coordination [14], adaptive edge-aware semantic interaction [21], and progressive feature interleaved fusion [23]. We admire this progress but are confused about the necessity. Is there a simple method that can perform comparably well? In this study, we seek to develop a simpler model to provide great performance in ORSI-SOD.

Recently, masked autoencoder (MAE) [25] has achieved great success in visual representation learning and delivered

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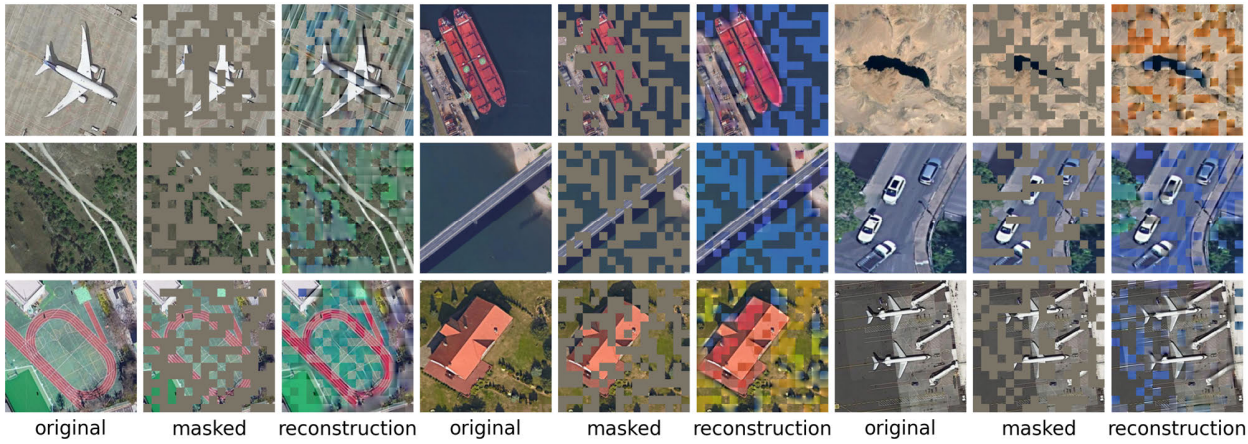


Fig. 1. Reconstruction results on the ORSI-4199 test set using an MAE trained on its training set. From left to right, the masking ratios are 0.6, 0.5, and 0.4.

promising potential in many downstream vision tasks, such as object detection and semantic segmentation. Witnessing the recent success of MAE in computer vision, we adapt MAE to ORSI-SOD and hope to improve the performance of a simple SOD model through MAE pretraining. The traditional way of MAE pretraining usually pretrains the encoder via masked image reconstruction and then transfers the parameters of the encoder to the downstream vision network for fine-tuning. However, due to the lack of a saliency supervision signal, almost no saliency information can be learned from the process of masked image reconstruction, which means that the SOD model can hardly obtain performance gain from the traditional MAE pretraining.

Considering that MAE has the ability to generate endless reconstructed images that are similar to the original image, we rethink MAE for ORSI-SOD from a pseudo image pretraining perspective and address the problem of limited transfer learning performance of MAE on SOD via a data-driven method. In this study, we integrate the SOD model and saliency supervision in MAE and propose a simple and effective framework for dynamic hybrid masking MAE (DHMMAE), utilizing MAE with dynamic hybrid masking (DHM) ratios to pretrain the SOD model. Specifically, we treat MAE as an online data augmenter to generate endless pseudo images for the SOD model to predict saliency maps, where saliency supervision is employed to ensure the model can learn robust saliency prior knowledge from the reconstructed images. In this way, the DHMMAE pretrained SOD model can achieve higher fine-tuning performance. Fig. 1 shows the reconstruction results of MAE on ORSI-4199 [37] test images, as can be observed, most of the masked salient objects (such as the airplane, the road, the playground, the ship, the bridge, the building, the lake, and the vehicle) are restored, and most of them are visually distinguishable compared with the masked image, which is beneficial for pseudo image pretraining. Besides, the poorly restored background increases the chaos level of the image, which can drive the SOD model to be more robust via pseudo image pretraining.

Apart from that, we also propose a simple and novel network, EANet, driven by DHMMAE pretraining for ORSI-SOD. Two key modules are designed for EANet to further enhance the model performance: the enhanced diverse feature aggregation module (EDFAM) and the adjacent-context shuffle spatial attention module (ASSAM). EDFAM aggregates diverse features via three different types of convolution layers and enhances them with a convolution block attention module (CBAM) [48]. ASSAM captures spatial location information of salient objects by employing channel shuffle operation and weighted spatial attention mechanism on the fused adjacent context. In this way, the proposed EANet driven by DHMMAE pretraining can generate saliency maps with richer local details and accurate salient locations.

Our key contributions are summarized as follows.

1) We present a simple and effective pretraining framework DHMMAE for ORSI-SOD, which significantly boosts the fine-tuning performance of SOD. As far as we know, this study marks the first application of MAE in ORSI-SOD.
2) We present EANet driven by DHMMAE pretraining for ORSI-SOD, and two key modules are designed to further improve the model performance. EDFAM can aggregate diverse features to provide richer local details, while ASSAM can capture accurate spatial locations of salient objects.
3) Experiments on three ORSI-SOD datasets demonstrate that our proposed method outperforms the cutting-edge methods. Ablation experiments are also conducted to validate the effectiveness of each component.

II. Related Work

A. SOD in ORSIs

ORSI-SOD plays a significant role in ORSI analysis [12], which is a research hotspot in the SOD community and recently numerous CNN-based and ViT-based methods are proposed [19]. Li et al. [16] proposed a nested network with a two-stream pyramid structure and the first dataset ORSSD for ORSI-SOD, which pioneers the field of ORSI-SOD. Zhang et al. [13] presented DAFNet with dense attention fluid and the extended EORSSD dataset for ORSI-SOD, which further promoted the development of this field. Cong et al. [18] proposed an RRNet with parallel multiscale attention

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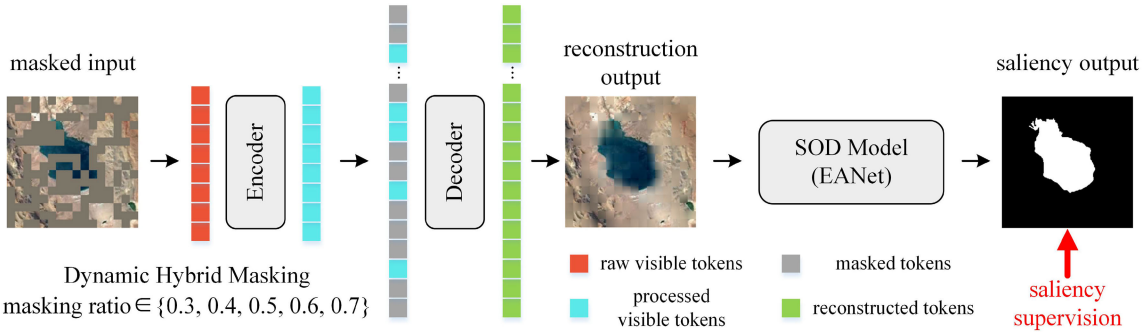


Fig. 2. Architecture of our DHMMAE, which consists of two parts: 1) masked image reconstruction and 2) saliency prediction from the reconstructed image (the predicted saliency map is supervised by ground truth). Specifically, we first pretrain the SOD model in DHMMAE via pseudo image pretraining. Then, we initialize the SOD model with the parameters obtained from the pretraining stage and fine-tune it for ORSI-SOD.

for ORSI-SOD, which effectively improved the accuracy and completeness of the prediction results. Li et al. [14] proposed ACCoNet, utilizing the coordination of adjacent context to improve the performance of ORSI-SOD, which can generate more accurate saliency maps.

Huang et al. [20] proposed PCDNet for ORSI-SOD, which adaptively captures context information for saliency detection. To address various interference issues present in ORSI, Zeng et al. [21] proposed AESINet for ORSI-SOD. To improve the local details and the integrity of saliency maps, Feng et al. [22] proposed a boundary-semantic collaborative guidance network for ORSI-SOD. To preserve detailed boundaries of prediction results and improve computational efficiency, Han et al. [23] proposed a progressive feature interleaved fusion network for ORSI-SOD. Li et al. [12] proposed a novel GeleNet based on ViT for SOD in ORSIs, which follows the global-to-local paradigm. Gong et al. [24] proposed an edge and skeleton guidance network for ORSI-SOD, which focuses on fully exploiting edge and skeleton information.

Although these methods have achieved great performance, they all focus on elaborately designing the sophisticated neural network architecture to improve the performance, which is a tiresome process that requires a large amount of trials and errors. We admire these excellent works but are confused about the necessity. Is there a simple method that can perform comparably well?

Witnessing the recent success of MAE, in this study, we seek to adopt MAE to enhance the performance of the SOD model and simplify the model design. Specifically, since MAE has the ability to automatically generate endless reconstructed images that are similar to the original image, we employ MAE to pretrain the SOD model by predicting saliency maps from the reconstructed images, where robust saliency prior knowledge can be learned from the sufficient pseudo images.

B. Masked Autoencoders

MAE, as a representative method of masked image modeling, learns latent visual representation via reconstructing the corrupted image from the visible image patches [26], which has shown promising transfer learning ability on many downstream vision tasks [27]. Recently, significant progress has been made in advancing MAE. He et al. [25] demonstrated that MAEs are scalable self-supervised learners for vision tasks, pioneering the development of MAEs. Gao et al. [28] presented ConvMAE using multiscale hybrid convolution-transformer architectures to further enhance the performance of MAE. To apply MAE to real-world applications, Ren et al. [29] explored knowledge distillation to transfer the success of large pretrained models to smaller ones. Liu et al. [30] proposed MixMAE for efficient pretraining of hierarchical vision transformers. Woo et al. [31] codesigned and scaled ConvNets with MAEs, which is the first work of a pure ConvNet-based MAE. Dong et al. [32] introduced BootMAE for pretraining a vision BERT model, which improves the original MAE with two core designs: momentum encoder and target-aware decoder.

Though these methods have achieved excellent performance, they mainly focus on elaborately designing MAE for universal visual representation learning, while ignoring how to adapt MAE to specific downstream vision tasks (e.g., ORSI-SOD in this research) to enhance the model performance. In this study, we explore the application of MAE in ORSI-SOD and seek to achieve better transfer learning performance.

III. Proposed Method

We present a simple and novel network, EANet, driven by DHMMAE pretraining for ORSI-SOD. Section III-A clarifies the framework overview. Section III-B presents the details of the DHMMAE. Sections III-C–III-E present key modules of EANet: EDFAM, ASSAM, and the decoder, respectively. Section III-F introduces the loss function.

A. Framework Overview

As illustrated in Figs. 2 and 3, our proposed approach consists of two stages: the pretraining stage (DHMMAE) and the fine-tuning stage (EANet). In the pretraining stage, we employ MAE (pretrained on the training data of EORSSD and ORSI-4199) to generate pseudo images for the SOD model to predict saliency maps; meanwhile, the saliency supervision is employed to ensure the SOD model can learn robust saliency prior knowledge from the abundant reconstructed images. Note that, we adopt the DHM strategy to provide input images with different masking ratios for MAE, which can increase the diversity of masked images and thus enhance the model's

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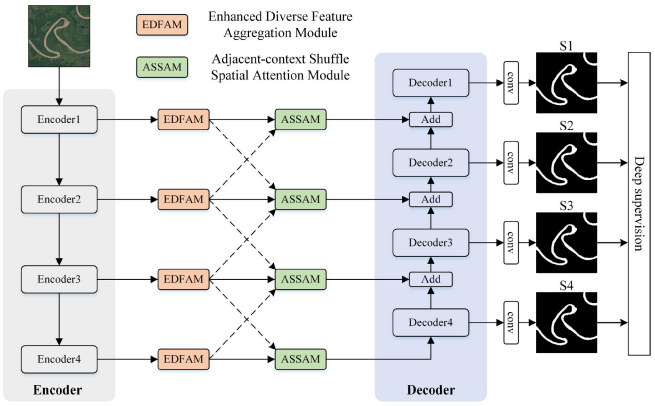


Fig. 3. Framework of our EANet, which consists of four major parts: encoder, EDFAM, ASSAM, and the decoder. We use the encoder to extract multilevel features at four scales and then feed the multiscale features to EDFAM to aggregate the diverse features and enhance them through the channel and spatial attention mechanism. Later, the adjacent multiscale features from EDFAM are fed to ASSAM for the mining of the spatial location of salient objects. After that, we use a simple decoder to fuse the multilevel features to predict the saliency maps.

robustness. In this way, the downstream SOD model can learn robust saliency information from the abundant random pseudo images, which can further enhance the fine-tuning performance.

In the fine-tuning stage, we first transfer the parameters to EANet from DHMMAE, which enables EANet to inherit the robust saliency-aware ability learned from the pseudo image pretraining stage. For EANet, we use the encoder (ResNet50 [20]) to extract multiscale features from the input image and then feed the multiscale features to EDFAM to aggregate the diverse features and enhance them through channel and spatial attention mechanisms. Later, the adjacent multiscale features from EDFAM are fed to ASSAM for the mining of the spatial location of salient objects via employing a weighted spatial attention mechanism on the fused adjacent feature. After that, we use a simple decoder to fuse the multilevel features to predict the saliency maps. Note that, we employ a deep supervision strategy, applying supervision to predicted saliency maps from all decoder blocks during training, which helps in refining the saliency predictions at various stages of the network, leading to improved accuracy.

In general, our EANet employs an encoder–decoder architecture that operates on multiscale features. This design allows the network to capture both local and global context, which is crucial for detecting salient objects of varying sizes and levels of complexity within an image.

B. DHM MAE

As shown in Fig. 2, we propose DHMMAE to pretrain the SOD model, and it takes the masked images with different masking ratios as inputs and outputs the corresponding reconstructed images for the SOD model to predict saliency maps, where the saliency supervision is employed at the end of the framework to ensure the SOD model can learn robust saliency prior knowledge from the sufficient pseudo images. Given that X_m is the masked input image, we first feed X_m to MAE to get the reconstruction output X_{rec}. Later, we feed X_{rec} to the SOD model to obtain the corresponding saliency output X_{sal}. The calculation process of DHMMAE is defined as follows:

$$X_{\text{rec}} = \text{MAE}(X_m) \tag{1}$$

$$X_{\text{sal}} = \text{SODModel}(X_{\text{rec}}). \tag{2}$$

Note that, MAE only works as an online data augmenter for the SOD model, which is frozen during training. Only the parameters of the SOD model are updated during the backward propagation process. In this case, we only calculate the loss of saliency output X_{sal}, details are described in Section III-F.

Algorithm 1 provides the pseudocode of DHMMAE in the pretraining stage. As can be seen, DHM employs a dynamic masking ratio list during pseudo image pretraining, which attaches different masking ratios to different input images according to their iteration number. Meanwhile, the masking ratio list updates after each pretraining epoch, where every element in the list moves forward in a cycle. During the pseudo image pretraining, the same input data in different epochs will be assigned with different masking ratios, which is beneficial to generate diverse reconstructed images from one input image with different masking ratios. In this way, the SOD model can learn robust saliency prior knowledge from various pseudo images, which can further enhance the SOD fine-tuning performance. Algorithm 2 presents the pseudocode of EANet in the fine-tuning stage.

Algorithm 1 Pseudocode of DHMMAE in a PyTorch-Like Style (Pretraining Stage)

```
def mrs_loop(mrs): # generate dynamic mask ratios
    mrs_next=[mrs[i] for i in range(1, len(mrs))]
    mrs_next.append(mrs[0])
    return mrs_next
mask_ratios=[0.3, 0.4, 0.5, 0.6, 0.7]
MAE.load_state_dict(pretrained_MAE)  # load pretrained MAE
MAE.eval()
SODModel.train()
for epoch in range(0, epochs):
    for i, (images, gts) in enumterate(data_loader): # train for one epoch
        cur_mr=mask_ratios[i%len(mask_ratios)]
        optimizer.zero_grad()
        _, pred, mask= MAE(imgs=images, mask_ratio= cur_mr)
        # 1 is removing, 0 is keeping, im_rec: reconstructed image
        im_rec = images * (1 - mask) + pred * mask
        sal = SODModel(im_rec)
        loss = BCE_Loss(sal, gts) + IOU_Loss(sal, gts)
        loss.backward()
        optimizer.step()
        scheduler.step()
    mask_ratios=mrs_loop(mask_ratios) # mask ratios update per epoch
```

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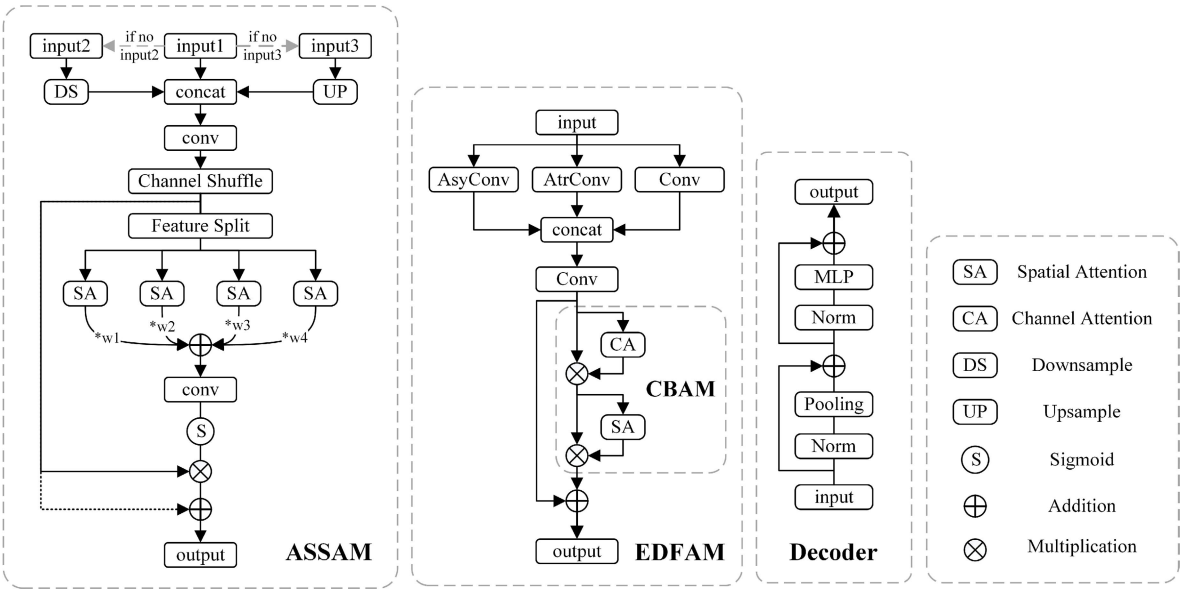


Fig. 4. Key modules of the proposed EANet: ASSAM, EDFAM, and the decoder. EDFAM aggregates diverse features via three different types of convolution layers and enhances them by CBAM. ASSAM captures spatial location information of salient objects by employing channel shuffle operation and weighted spatial attention mechanism on the fused adjacent context. The decoder is a simplified PoolFormer block, which is used to fuse the multilevel features and generate the saliency map.

Algorithm 2 Pseudocode of EANet in a PyTorch-Like Style (Fine-Tuning Stage)

```
# load pretrained EANet from DHMMAE
model.load_state_dict(pretrained_EANet)
for epoch in range(epochs):
    model.train()
    for (imgs, gts) in data_loader: # train for one epoch
        optimizer.zero_grad()
        S1, S2, S3, S4 = model(imgs)
        loss1 = BCE_Loss(S1, gts) + IOU_Loss(S1, gts)
        loss2 = BCE_Loss(S2, gts) + IOU_Loss(S2, gts)
        loss3 = BCE_Loss(S3, gts) + IOU_Loss(S3, gts)
        loss4 = BCE_Loss(S4, gts) + IOU_Loss(S4, gts)
        loss = loss1 + loss2/2 + loss3/4 + loss4/8
        loss.backward()
        optimizer.step()
        scheduler.step()
```

C. Enhanced Diverse Feature Aggregation Module

As shown in Fig. 3, two key modules are proposed to further enhance the performance of EANet: EDFAM and ASSAM. Inspired by the work of ICON [33], which designed the diverse feature aggregation (DFA) module and incorporated three kinds of convolution blocks with different kernel sizes and shapes to deal with the shape diversity of different objects: asymmetric convolution [34], atrous convolution [35], and original convolution operations. We further mine the advantages of DFA and enhance it with channel and spatial attention mechanisms.

Our EDFAM aggregates features from different convolution layers. By doing so, it enhances the model's ability to capture diverse shapes and textures present in salient objects. The use of channel and spatial attention mechanisms within EDFAM further refines the feature representation, allowing the model to focus on the most relevant features for saliency prediction.

As shown in Fig. 4, we design EDFAM, an improved DFA, which adopts channel attention and spatial attention to my saliency information from the aggregated diverse feature. Besides, we also employ a skip connection to reduce the model complexity and help the model converge quickly. The overall procedure of EDFAM can be defined as follows:

$$X_{dfa} = \mathrm{Conv}\big(\mathrm{Concat}\big[\mathrm{Asy}(X), \mathrm{Atr}(X), \mathrm{Conv}(X)\big]\big) \tag{3}$$

$$X_{ca} = X_{dfa} \otimes \mathrm{CA}(X_{dfa}) \tag{4}$$

$$X_{e\mathrm{out}} = X_{ca} \otimes \mathrm{SA}(X_{ca}) + X_{dfa} \tag{5}$$

where Asy, Atr, and Conv denote asymmetric convolution, atrous convolution, and the original convolution, respectively. Concat is the concatenation operation. CA and SA are channel attention and spatial attention, respectively. Concretely, CA is implemented by an adaptive average pooling, a Conv layer with a sigmoid function; and SA is implemented by a Conv layer followed by a sigmoid function. $X_{e\mathrm{out}}$ is the output of EDFAM.

D. Adjacent-Context Shuffle Spatial Attention Module

As shown in Fig. 3, ASSAM is employed after EDFAM to detect the spatial locations of salient objects. Inspired by the work of GeleNet [12], which designed the shuffle weighted spatial attention module (SWSAM) to enhance salient areas across different channels. Specifically, the original SWSAM first produces the local spatial information from each subfeature and then employs the weighted fusion operation to generate the final spatial information for location enhancement.

However, the original SWSAM only operates on the single-scale feature from one encoder stage, which has a

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limited effective receptive field when applying SWSAM to the CNN encoder. To address this problem, we proposed ASSAM, which is designed to capture spatial relationships between salient objects and their surroundings. It uses a channel shuffle operation to combine features from adjacent scales, followed by a weighted spatial attention mechanism. This process helps in accurately localizing the spatial extent of salient objects by considering their contextual relationships within the image.

Concretely, we first fuse the multiscale features from adjacent encoder stages and then employ channel shuffle operation and weighted spatial attention mechanism on the fused adjacent multiscale feature, which expands the effective receptive field and thus improves the detection accuracy. The complete process of ASSAM is defined as follows:

$$X_{amf} = \text{Conv}(\text{Concat}[\text{DS}(X2), X1, \text{UP}(X3)]) \quad (6)$$

$$X_{cs} = \text{channel_shuffle}(X_{amf}) \quad (7)$$

$$x1, x2, x3, x4 = \text{split}(X_{cs}) \quad (8)$$

$$xi = \text{SA}(xi), \quad i = 1, 2, 3, 4 \quad (9)$$

$$X_{ws} = \text{sigmoid}\left(\text{Conv}\left(\sum_{n=1}^{4} wi * xi\right)\right) \quad (10)$$

$$X_{a\text{out}} = X_{cs} \otimes X_{ws} + X_{cs} \quad (11)$$

where $X1$, $X2$, and $X3$ are the features from the current, the previous, and the latter encoder stages, respectively; DS and UP are downsample and upsample operations, respectively; sigmoid is the sigmoid activation function; and $X_{a\text{out}}$ is the output of ASSAM.

E. Decoder

As shown in Figs. 3 and 4, we adopt a simple transformer-style decoder for EANet. Inspired by the work of MetaFormer [36], which proved that the general architecture of the transformers, instead of the specific token mixer module, is more essential to the model's performance. To make the SOD model simple, we use the simple PoolFormer [36] block as the decoder of our proposed EANet. Specifically, in the decoder of EANet, we replace the attention module in the transformer with a lightweight pooling operation and set the expansion rate in MLP to 1 to reduce computation. This design choice ensures that the model remains efficient in terms of computational resources and inference speed. The overall procedure of the decoder can be defined as follows:

$$X_P = \text{Pooling}(\text{Norm}(X)) + X \quad (12)$$

$$X_{d\text{out}} = \text{MLP}(\text{Norm}(X_P)) + X_P \quad (13)$$

where Norm is the group normalization operation, MLP is a multilayer perceptron, and Pooling is implemented by "nn.AvgPool2d" in PyTorch.

F. Loss Function

As shown in the right part of Fig. 3, we add a Conv layer after each decoder block to produce four saliency maps. S1 is the final prediction and the other three saliency maps (S2, S3, and S4) are used for auxiliary training. The binary cross-entropy (BCE) loss and the intersection-over-union (IoU) loss are employed in the loss function for the training of EANet, which can be written as follows:

$$L = \sum_{i=1}^{4} \frac{1}{2^{i-1}} (L_{\text{bce}}(\text{Si}, \text{GT}) + L_{\text{iou}}(\text{Si}, \text{GT})) \quad (14)$$

where L_{bce} and L_{iou} are the BCE loss and IoU loss, respectively. GT is the ground truth.

IV. Experiments

A. Benchmark Dataset

We conduct experiments on three benchmark ORSI-SOD datasets: ORSSD [16], EORSSD [13], and ORSI-4199 [37].

ORSSD is the first public ORSI-SOD dataset. It has 800 images with pixel-wise annotations, where the training part has 600 images and the testing part has 200 images. EORSSD has 2000 images with pixel-wise annotations for ORSI-SOD, where the training section contains 1400 images and the testing section involves 600 images. ORSI-4199 is the biggest public ORSI-SOD dataset, and it has 4199 images with pixel-wise annotations, where the training part involves 2000 images and the testing part contains 2199 images.

B. Evaluation Metrics

We adopt four widely used evaluation metrics to assess the model performance, including: 1) F-measure; 2) mean absolute error (MAE); 3) S-measure; and 4) E-measure.

F-measure (F_β) takes both precision and recall into account, which is defined as

$$F_\beta = \frac{(1 + \beta^2) j \times \text{Precision} \times \text{Recall}}{\beta^2 \times \text{Precision} + \text{Recall}} \quad (15)$$

where β^2 is set to 0.3, as suggested in previous work [20], [21].

MAE represents the average per-pixel difference between prediction and ground truth, and it is defined as

$$\text{MAE} = \frac{1}{H \times W} \sum_{r=1}^{H} \sum_{C=1}^{W} |P(r, c) - \text{GT}(r, c)| \quad (16)$$

where P is the prediction and GT is the ground truth. (H, W) and (r, c) are (height, width) and pixel coordinates, respectively.

S-measure (S_m) computes the structure similarity of region-aware S_r and the object-aware S_o, which is defined as

$$S_m = (1 - \alpha) S_r + \alpha S_o \quad (17)$$

where α is usually set to 0.5.

E-measure (E_m) can assess the structural similarity between prediction and ground truth, and it is defined as

$$E_m = \frac{1}{W \times H} \sum_{i=1}^{W} \sum_{j=1}^{H} \varphi_P(i, j) \quad (18)$$

where φ_P is the enhanced alignment matrix.

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TABLE I

QUANTITATIVE COMPARISONS. THE TOP TWO RESULTS ARE IN RED AND BLUE, RESPECTIVELY. ↑ MEANS THE BIGGER THE BETTER, WHILE ↓ THE OPPOSITE. THE $-V$ AND $-R$ INDICATE WHETHER THE ENCODER OF THE MODEL IS VGG OR RESNET

Methods	Year	ORSSD				EORSSD				ORSI-4199			
		F_β↑	MAE↓	S_m↑	E_m↑	F_β↑	MAE↓	S_m↑	E_m↑	F_β↑	MAE↓	S_m↑	E_m↑
R3Net [38]	2018	.7456	.0399	.8141	.8913	.7498	.0171	.8184	.9483	.7847	.0401	.8142	.8880
PoolNet [39]	2019	.7706	.0358	.8403	.9343	7545	.0210	.8207	.9292	.8010	.0541	.8271	.8964
EGNet [40]	2019	.8332	.0216	.8721	.9731	.7880	.0110	.8601	.9570	.8267	.0440	.8464	.9161
LVNet [13]	2019	.8263	.0207	.8815	.9456	.7794	.0146	.8630	.9254	-	-	-	-
MINet [41]	2020	.8761	.0144	.9040	.9545	.8344	.0093	.9040	.9442	.7988	.0504	.8116	.8961
GateNet [42]	2020	.8871	.0137	.9186	.9664	.8566	.0095	.9114	.9610	.8450	.0393	.8545	.9256
F3Net [49]	2020	.8502	.0203	.8874	.9327	.8410	.0129	.8923	.9374	.7519	.0607	.8005	.8519
GCPANet [50]	2020	.8482	.0203	.8743	.9351	.8127	.0137	.8557	.9280	.7974	.0526	.8298	.8860
HVPNet [43]	2021	.7938	.0255	.8610	.9320	.8036	.0110	.8734	.9482	.8295	.0419	.8471	.9201
PA-KRN [44]	2021	.8890	.0139	.9239	.9680	.8639	.0104	.9192	.9616	.8415	.0382	.8491	.9280
DPORTNet [45]	2022	.8309	.0220	.8827	.9214	.8363	.0150	.8960	.9423	.7789	.0569	.8094	.8759
DNTD [46]	2022	.8231	.0217	.8698	.9286	.8189	.0113	.8957	.9378	.8310	.0425	.8444	.9158
PGNet [51]	2022	.8928	.0150	.9114	.9595	.8665	.0103	.9055	.9494	.8178	.0450	.8395	.8990
MJRBM [37]	2022	.8842	.0163	.9204	.9623	.8656	.0099	.9197	.9646	.8493	.0374	.8593	**.9311**
ICON [33]	2023	.8939	**.0116**	**.9256**	.9704	.8622	**.0073**	.9185	.9687	**.8763**	**.0282**	**.8752**	**.9521**
ERPNet-R [47]	2023	.8974	.0135	.9254	.9710	**.8743**	.0082	**.9252**	.9665	.8528	.0388	.8636	.9292
GeleNet-V [12]	2023	**.9023**	.0130	.9252	**.9744**	.8721	**.0080**	.9241	**.9723**	.8444	.0391	.8540	.9283
EANet	**Ours**	**.9180**	**.0113**	**.9310**	**.9759**	**.8974**	**.0073**	**.9286**	**.9740**	**.8563**	**.0361**	**.8683**	.9287

C. Implementation Details

We adopt the PyTorch framework to implement the proposed method and use an NVIDIA RTX 3090 GPU for training acceleration. We conduct all the experiments following data augmentation as recommended in previous work [13], [16], [37]. We first pretrain MAE-base to the training data of EORSSD and ORSI-4199 for 300 epochs via mask image reconstruction following the same training strategy as MAE [25] (initial learning rate: $1e^{-3}$, masking ratio: 0.75, input size: 256 × 256, batch size: 64, and weight decay: 0.05). Then, we transfer the parameters of the MAE-base to DHMMAE to pretrain SOD model.

For the pretraining stage, the pretraining data are the training set of the corresponding ORSI-SOD dataset. We adopt MAE-base as the reconstruction model and EANet (ResNet50 [20] as the encoder) as the SOD model in DHMMAE. DHMMAE is pretrained using the Adam optimizer for 100 epochs (batch size: 32, initial learning rate: $1e^{-3}$, and input size: 256 × 256), and the "polyLR" scheduler is also employed.

For the fine-tuning stage, our EANet adopts ResNet50 as the encoder and initializes with the parameters of the SOD model from the DHMMAE pretraining stage. Our EANet is trained using the Adam optimizer for 50 epochs (batch size: 32, max learning rate: $1e^{-3}$, and input size: 352 × 352), and the "polyLR" scheduler is also employed.

In DHMMAE, MAE and SOD models are trained in stages. Specifically, we first train MAE on the training data of EORSSD and ORSI-4199 for 300 epochs via masked image reconstruction, and this process makes MAE act as an online data augmenter to generate diverse pseudo images during the pretraining phase, while the SOD model is pretrained based on these pseudo images. In DHMMAE pretraining, MAE is frozen, and only the parameters of the SOD model are updated. This staged training mechanism enables the SOD model to learn rich saliency priors from the various pseudo images generated by MAE, thereby improving its performance.

D. Comparisons With State of the Arts

We compare our proposed method with 17 state-of-the-art SOD methods, including R3Net [38], PoolNet [39], EGNet [40], LVNet [13], MINet [41], GateNet [42], F3Net [49], GCPANet [50], HVPNet [43], PA-KRN [44], DPORTNet [45], DNTD [46], PGNet [51], MJRBM [37], ICON [33], ERPNet [47], and GeleNet [12]. All of the results for the compared methods are obtained from public benchmarks [12], [16], [21], [37] or by running the released codes from the authors. For the method without code access, we cannot measure their real performance on the ORSI-SOD dataset, so they are marked as—in Table I.

1) Quantitative Comparison: Table I shows the quantitative comparison of three ORSI-SOD datasets, and it can be seen that our EANet achieves the highest score across all metrics on ORSSD and EORSSD while achieving the competitive score on ORSI-4199, which also demonstrates the superiority of our proposed method.

For ORSSD, our EANet achieves the best results on four metrics. Specifically, our EANet reaches 0.9180 and 0.9310 on F_β and S_m, which exceeds the second-best method (GeleNet-V and ICON) by 1.57% and 0.54%.

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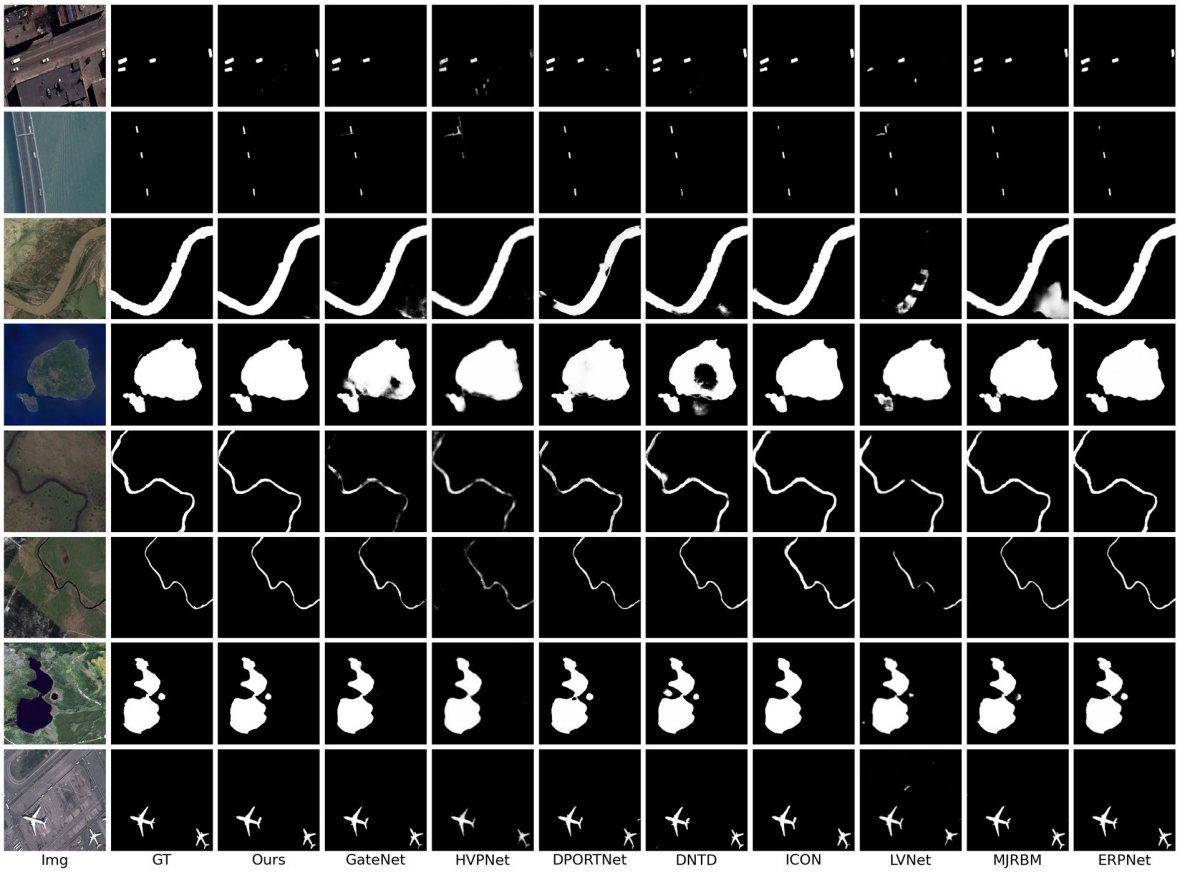


Fig. 5. Visual comparisons on EORSSD testing data. Multiple tiny salient objects (1st, 2nd row), low contrast between foreground and background (3rd, 4th row), salient object with irregular geometry shape (5th, 6th row), and multiscale salient objects (7th, 8th row).

For EORSSD, our EANet also achieves the best results on four metrics. Specifically, our EANet reaches 0.8974 and 0.9286 on F_β and S_m, which surpasses the second-best method (ERPNet-R) by 2.31% and 0.34%.

For ORSI-4199, our EANet achieves the second-best results on F_β, MAE, and S_m, and a competitive result on E_m. Although ICON surpasses our method on ORSI-4199, it has a sophisticated model design, while our EANet is quite simple. Besides, EANet is the first application of MAE on ORSI-SOD and there is still a large room for improvement, which will motivate further research.

In general, with the help of DHMMAE pseudo image pretraining, our simple EANet achieves the best results on ORSSD and EORSSD while maintaining competitive results on ORSI-4199, which also exhibits the excellent performance of the overall framework.

2) Visual Comparison: The efficacy of our proposed EANet is vividly demonstrated in Fig. 5, where it can predict more accurate and complete saliency maps compared with eight competing methods on the EORSSD. Our method's superior performance is evident across a spectrum of challenging scenarios, e.g., multiple tiny salient objects (1st, 2nd row), low contrast between foreground and background (3rd, 4th row), salient object with irregular geometry shape (5th, 6th row), and multiscale salient objects (7th, 8th row). Overall, the visual comparisons validate the effectiveness of our EANet in generating saliency maps that are not only accurate but also free from noise. The model's integral detection capability and its consistent performance across diverse and challenging conditions highlight the robustness and reliability of our approach.

3) Comparison on Model Complexity: We report the model parameters, floating point operations (FLOPs), and inference speed (frames per second, FPS) in Table II. As can be observed, our simple EANet has a relatively low model parameter of 25.2 M, which is fairly lightweight compared with most methods. Besides, our EANet has FLOPs of 30.1 G, which is also lower than most of the compared methods. In addition, our EANet reaches the real-time inference speed of 63.5 FPS, which is the second-fastest method among the compared methods. These measurements also demonstrate that our EANet has the advantage in model complexity compared with other competing methods.

E. Ablation Studies

In this section, we conduct thorough ablation experiments to verify the effectiveness of our proposed method.

1) Ablation Study on Performance of DHMMAE With Different Masking Ratios: To fully mine the potential of MAE pseudo image pretraining and select the best optimal masking strategy, we pretrain DHMMAE with different masking strategies and then fine-tune EANet to report the test results on EORSSD. Since smaller masking ratios (such as 0.1 and 0.2) usually fail to mask salient objects while bigger masking

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TABLE II
COMPARISON ON MODEL COMPLEXITY

Methods	Input Size	Param (M) ↓	FLOPs (G) ↓	Speed (FPS) ↑
R3Net [38]	300×300	56.2	47.5	2
PoolNet [39]	256×256	52.5	97.6	80.9
EGNet [40]	256×256	108.1	313.9	43.5
MINet [41]	320×320	47.6	146.3	12
GateNet [42]	256×256	100.0	96.3	51.5
F3Net [49]	256×256	25.5	8.7	60.8
GCPANet [50]	256×256	67.1	34.8	59.5
PGNet [51]	256×256	72.7	18.4	33.3
MJRBM [37]	256×256	43.5	101.3	15.5
ICON [33]	256×256	92.1	52.6	37
ERPNet [47]	256×256	77.2	343.2	32.5
GeleNet [12]	256×256	25.5	6.2	51.2
EANet	256×256	25.2	30.1	63.5

TABLE III
ABLATION STUDY ON PERFORMANCE OF DHMMAE WITH DIFFERENT MASKING RATIOS

Masking Ratio	EORSSD			
	F_β↑	MAE↓	S_m↑	E_m↑
0.3	.8850	.0089	.9165	.9653
0.4	.8875	.0081	.9182	.9661
0.5	.8855	.0086	.9185	.9669
0.6	.8842	.0085	.9176	.9696
0.7	.8842	.0085	.9167	.9650
DHM	**.8974**	**.0073**	**.9286**	**.9740**

ratios (such as 0.8 and 0.9) will lead to poor reconstruction results (almost no masked patches can be restored), we select 0.3, 0.4, 0.5, 0.6, and 0.7 as the testing masking ratios in this ablation study.

As can be observed in Table III, our proposed DHM delivers top performance across all metrics on the EORSSD dataset, which is obviously better than other masking strategies. DHM uses a dynamic masking ratio strategy during pseudo image pretraining, which can generate more diverse reconstructed images compared with the single masking ratio strategy. In this way, the SOD model can learn more robust saliency prior knowledge from various pseudo images, which can further enhance fine-tuning performance.

2) Ablation Study on Influence of the Pretraining Stage: To verify the effectiveness of the pretraining stage, we train EANet with and without DHMMAE pretraining. Table IV reports the results on the three ORSI-SOD datasets, and it can be seen that the pretrained model performs better than the unpretrained one under all evaluation metrics. Specifically, in terms of the S_m score, the pretrained model reaches 0.9310, 0.9286, and 0.8683 on the three datasets, and it exceeds the unpretrained one by 1.96%, 2.01%, and 0.95%, which improvement is significant.

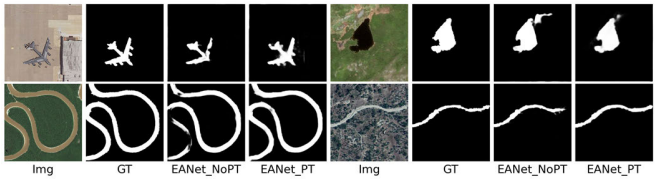


Fig. 6. Visual comparisons of EANet under different cases: pretrained or not. Specifically, EANet_NoPT denotes the EANet without pretraining stage, and EANet_PT denotes the DHMMAE pretrained EANet.

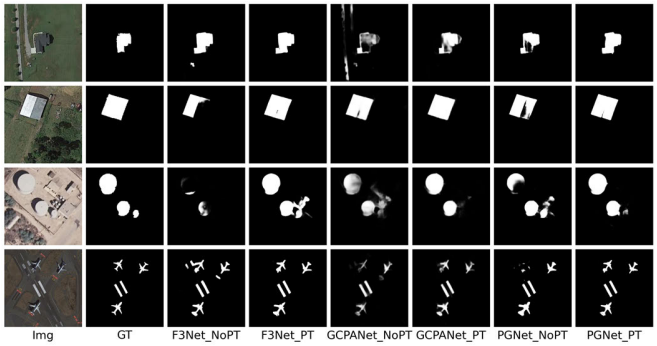


Fig. 7. Visual comparisons of other SOD methods under different cases: pretrained or not. Specifically, _NoPT denotes the method without pretraining, and _PT denotes the method pretrained by DHMMAE.

Fig. 6 presents the visual comparisons of EANet with or without pretraining stage, and we can observe that after pretrained by DHMMAE, the quality of the predicted saliency maps gets improved a lot, which is much more accurate and complete compared with the unpretrained one. The above ablation results prove that DHMMAE can significantly improve the performance of our EANet.

3) Ablation Study on Validity of DHMMAE on Other SOD Methods: To verify the generalization performance of the proposed DHMMAE on other SOD methods, we apply DHMMAE to three mainstream SOD methods: F3Net [49], GCPANet [50], and PGNet [51], and report the comparative results on the three datasets.

As can be observed in Table V, for the three methods, the pretrained model is obviously better than the unpretrained one under all metrics on the three datasets. For instance, for GCPANet, in terms of the S_m score, the pretrained model reaches 0.9118, 0.8856, and 0.8477 on the three datasets. Also, it surpasses the unpretrained one by 3.75%, 2.99%, and 1.79% on the three datasets, which performance increase is quite considerable.

Fig. 7 shows the visual comparisons of other SOD methods under different cases: pretrained or not. As can be seen, most of the pretrained models can generate more accurate saliency maps compared to the unpretrained ones, which shows that our proposed DHMMAE can also visually improve the performance of SOD methods. The abovementioned analysis comprehensively proves the validity of our DHMMAE.

4) Ablation Study on Contributions of Key Modules in EANet: To prevent the influence of the pretraining stage, we train the variants without DHMMAE pretraining in this ablation study. As shown in Table VI, independently adding every single module improves the performance of the network,

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TABLE IV
ABLATION STUDY ON INFLUENCE OF THE PRETRAINING STAGE. THE BEST RESULTS ARE IN BOLD

Methods	ORSSD				EORSSD				ORSI-4199			
	$F_\beta\uparrow$	MAE$\downarrow$	$S_m\uparrow$	$E_m\uparrow$	$F_\beta\uparrow$	MAE$\downarrow$	$S_m\uparrow$	$E_m\uparrow$	$F_\beta\uparrow$	MAE$\downarrow$	$S_m\uparrow$	$E_m\uparrow$
Pretrained	**.9180**	**.0113**	**.9310**	**.9759**	**.8974**	**.0073**	**.9286**	**.9740**	**.8563**	**.0361**	**.8683**	**.9287**
Un-Pretrained	.8970	.0153	.9114	.9655	.8732	.0096	.9085	.9622	.8350	.0403	.8588	.9084

TABLE V
ABLATION STUDY ON VALIDITY OF DHMMAE ON OTHER SOD METHODS. THE BEST RESULTS OF EACH CASE ARE IN BOLD

Methods	Type	ORSSD				EORSSD				ORSI-4199			
		$F_\beta\uparrow$	MAE$\downarrow$	$S_m\uparrow$	$E_m\uparrow$	$F_\beta\uparrow$	MAE$\downarrow$	$S_m\uparrow$	$E_m\uparrow$	$F_\beta\uparrow$	MAE$\downarrow$	$S_m\uparrow$	$E_m\uparrow$
F3Net	Un-Pretrained	.8502	.0203	.8874	.9327	.8410	.0129	.8923	.9374	.7519	.0607	.8005	.8519
	Pretrained	**.8969**	**.0137**	**.9155**	**.9578**	**.8854**	**.0085**	**.9160**	**.9596**	**.8090**	**.0453**	**.8434**	**.8938**
GCPANet	Un-Pretrained	.8482	.0203	.8743	.9351	.8127	.0137	.8557	.9280	.7974	.0526	.8298	.8860
	Pretrained	**.8988**	**.0152**	**.9118**	**.9659**	**.8531**	**.0110**	**.8856**	**.9474**	**.8169**	**.0439**	**.8477**	**.8973**
PGNet	Un-Pretrained	.8928	.0150	.9114	.9595	.8665	.0103	.9055	.9494	.8178	.0450	.8395	.8990
	Pretrained	**.9024**	**.0138**	**.9218**	**.9621**	**.8844**	**.0089**	**.9172**	**.9604**	**.8286**	**.0409**	**.8500**	**.9062**

TABLE VI
ABLATION STUDY ON CONTRIBUTIONS OF KEY MODULES IN EANET: EDFAM AND ASSAM. BASELINE IS THE ENCODER–DECODER NETWORK. THE BEST RESULTS ARE IN BOLD

No.	Baseline	EDFAM	ASSAM	ORSSD				EORSSD			
				$F_\beta\uparrow$	MAE$\downarrow$	$S_m\uparrow$	$E_m\uparrow$	$F_\beta\uparrow$	MAE$\downarrow$	$S_m\uparrow$	$E_m\uparrow$
1	✓			.8749	.0177	.9017	.9532	.8669	.0112	.9038	.9572
2	✓	✓		.8860	.0165	.9051	.9572	8708	.0104	.9068	.9594
3	✓		✓	.8871	.0162	.9072	.9613	.8719	.0107	.9080	.9617
4	✓	✓	✓	**.8970**	**.0153**	**.9114**	**.9655**	**.8732**	**.0096**	**.9085**	**.9622**

TABLE VII
ABLATION STUDY ON VALIDITY OF MODIFICATIONS IN EDFAM AND ASSAM. THE BEST RESULTS ARE IN BOLD

	Models	ORSSD				EORSSD			
		$F_\beta\uparrow$	MAE$\downarrow$	$S_m\uparrow$	$E_m\uparrow$	$F_\beta\uparrow$	MAE$\downarrow$	$S_m\uparrow$	$E_m\uparrow$
	EANet(Ours)	**.8970**	**.0153**	**.9114**	**.9655**	**.8732**	**.0096**	**.9085**	**.9622**
EDFAM	w/o CBAM	.8881	.0159	.9090	.9580	.8725	.0101	.9085	.9604
ASSAM	w/o ACF	.8950	.0165	.9083	.9635	.8729	.0104	.9070	.9601

which proves the effectiveness of each module of our EANet. When EDFAM and ASSAM are both added to the baseline, the network reaches its peak performance, which reaches 0.9114 and 0.9085 in terms of the S_m score on ORSSD and EORSSD, respectively. This also reveals the effectiveness of the overall architecture.

Apart from that, as displayed in Fig. 8, the baseline model performs badly, which fails to detect all salient objects or mistakes some background objects as the salient target. With the addition of EDFAM or ASSAM, the quality of the saliency map gets improved and a larger proportion of the salient objects is detected. When EDFAM and ASSAM are both added to the baseline, the saliency maps generated by our EANet are visually closest to GTs. The visual comparison and subsequent analysis conclusively demonstrate that the introduction of EDFAM and ASSAM is instrumental in elevating the performance of the baseline model. The synergistic effect of these two key modules endows EANet with the capability to produce high-fidelity saliency maps that accurately reflect the true salient content of the scene.

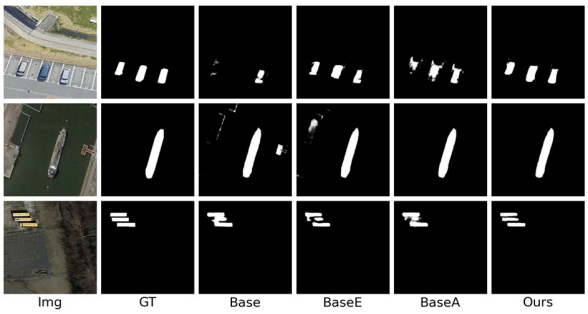


Fig. 8. Visual comparisons with different variants in Table VI. Base, BaseE, and BaseA are baseline, baseline + EDFAM, and baseline + ASSAM, respectively.

5) Ablation Study on Validity of Modifications in EDFAM and ASSAM: We introduce two variants of our model: 1) EDFAM without CBAM (w/o CBAM) and 2) ASSAM without adjacent-context feature (w/o ACF). As shown in Table VII, across all metrics and datasets, both variants

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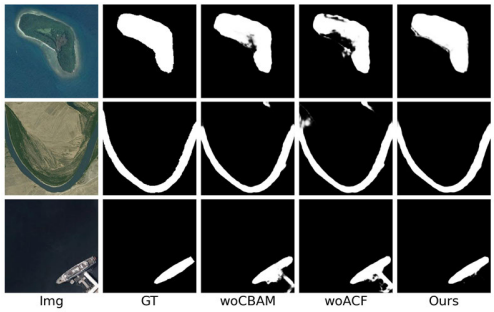


Fig. 9. Visual comparisons with different variants in Table VII. woCBAM and woACF are EDFAM (w/o CBAM) and ASSAM (w/o ACF), respectively.

exhibited diminished performance relative to our default EANet, which proves that both two modifications are necessary for the network to improve performance.

In addition, Fig. 9 presents the visual comparisons of different variants in this ablation study. As can be seen, both two variants perform badly, as saliency maps are incomplete and blurred. By contrast, with all components equipped, our default EANet predicts the saliency maps closest to GTs, which also confirms the effectiveness of the modifications in the two key modules.

V. Conclusion

In this study, we pioneer the application of MAE to ORSI-SOD, rethink MAE for ORSI-SOD from a pseudo image pretraining perspective, and propose a simple and effective SOD pretraining framework DHMMAE, which further boosts the performance of the SOD model. Moreover, we also design a simple and novel network, EANet, driven by DHMMAE pretraining for ORSI-SOD, and two key modules are proposed to improve the model performance, namely, EDFAM and ASSAM. Abundant experiments on three public ORSI-SOD datasets demonstrate that our proposed method outperforms the cutting-edge methods, which is also a milestone of MAE application in ORSI-SOD. We hope our approach will facilitate the development of SOD and MAE applications in downstream vision tasks and thus motivate further research.

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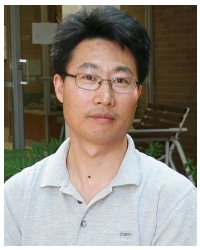
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